Automatismes

Série 6 — Évaluation — Corrigé

Q1

A1 · Comparer deux nombres

★ **C** : $\dfrac{5}{4} = 1{,}25$ est le plus petit ($\dfrac{126}{100} = 1{,}26$; $1{,}3 = 13 \times 10^{-1}$).

★★ $\dfrac{113}{100} > \dfrac{9}{8} > 11{,}2 \times 10^{-1} > 1{,}1$ car $1{,}13 > 1{,}125 > 1{,}12 > 1{,}1$.

★★★ $\dfrac{4}{7} > \dfrac{5}{9}$ car $\dfrac{4}{7} = \dfrac{36}{63}$ et $\dfrac{5}{9} = \dfrac{35}{63}$.

Q2

A13 · Proportions

★ **A** : $35 \times 240 \times 0{,}01$ ($35\,\%$ de 240, c'est $240 \times \dfrac{35}{100} = 84$).

★★ $450 \times 0{,}12$ ou $450 \times \dfrac{12}{100}$: $12\,\%$ de 450 vaut 54.

★★★ $64\,\%$ de $25 = 25\,\%$ de $64 = \dfrac{64}{4} = 16$.

Q3

A5 · Encadrer

★ $7 < \sqrt{57} < 8$ car $7^2 = 49 < 57 < 64 = 8^2$.

★★ $4 < \sqrt{20} < 5$ (car $16 < 20 < 25$) et $11 < \sqrt{130} < 12$ (car $121 < 130 < 144$).

★★★ $7{,}5 < \sqrt{57} < 7{,}6$ car $7{,}5^2 = 56{,}25 < 57 < 57{,}76 = 7{,}6^2$.

Q4

A4 · Écritures d'un nombre

★ $B = 0{,}7 + 0{,}36 = 1{,}06$.

★★ $C = 0{,}4 + 0{,}009 = 0{,}409$ et $D = 7{,}2 - 0{,}08 = 7{,}12$.

★★★ $\dfrac{3}{8} = 0{,}375$, donc $E = 1{,}45 + 0{,}375 = 1{,}825 = \dfrac{1\,825}{1\,000} = \dfrac{73}{40}$.

Q5

A9 · Développer, factoriser

★ $C = -3 \times 4x - 3 \times (-5) + 7 = -12x + 15 + 7 = -12x + 22$.

★★ $E = 4x - 6x + 15 = -2x + 15$.

★★★ $F = -5 + 3x + 2x^2 + 8x = 2x^2 + 11x - 5$.

Q6

A5 · Ordre de grandeur

★ **D** : 160 ($19\,\%$ de $812 \approx 20\,\%$ de $800 = 160$).

★★ $49\,\%$ de $6\,030 \approx 50\,\%$ de $6\,000 = 3\,000$; $11\,\%$ de $2\,980 \approx 10\,\%$ de $3\,000 = 300$.

★★★ Estimation : $79 \approx 80$ et $-32\,\% \approx -30\,\%$, donc $80 \times 0{,}7 = 56$ €. Prix exact : $79 \times 0{,}68 = 53{,}72$ €.

Q7

A10 · Équations

★ $-5x + 8 = -4 \iff -5x = -12 \iff x = \dfrac{12}{5} = 2{,}4$.

★★ $\dfrac{x}{3} + 4 = -2 \iff \dfrac{x}{3} = -6 \iff x = -18$.

★★★ $5 - \dfrac{3}{4}x = 11 \iff -\dfrac{3}{4}x = 6 \iff x = 6 \times \left(-\dfrac{4}{3}\right) = -8$.